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FLOOD CONTROL BARRIER SYSTEM

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in partial fulfillment of the course
P411 | EMBEDDED SYSTEM

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INTRODUCTION

The Flood Control Barrier System is an automated water-level monitoring and barrier control device designed to help reduce the risk of flooding in vulnerable areas. The system detects water levels using an analog water sensor, processes the data using an Arduino microcontroller, and automatically adjusts a servo-operated barrier while providing visual and audible warnings through LEDs, a buzzer, and an LCD display.

Flooding is one of the most frequent and destructive natural hazards worldwide. According to the United Nations Office for Disaster Risk Reduction (UNDRR), floods account for a significant portion of global weather-related disasters and are responsible for extensive economic and human losses, with annual average losses reaching billions of dollars globally (<https://www.undrr.org/gar/gar2025/hazard-exploration/floods>). Additionally, the World Bank estimates that approximately 1.81 billion people are currently exposed to significant flood risk events worldwide, especially in low- and middle income countries (<https://blogs.worldbank.org/climatechange/flood-risk-already-affects-181-billion-people-climate-change-and-unplanned>).

Because water levels can rise quickly during intense rainfall and storm events, early detection and automated response systems are crucial to minimize property damage and improve community safety. Flood control systems that integrate real-time monitoring and barrier mechanisms can provide a fast and effective means of protecting infrastructure and lives (https://www.who.int/health-topics/floods#tab=tab_1).

This project was developed to demonstrate how embedded systems can be applied to real-world disaster prevention. By combining automatic monitoring,

visual and audible warning indicators, and a movable flood barrier, the system offers a simple, affordable, and practical solution that can be implemented in small-scale flood-prone areas.

OBJECTIVES

To develop a reliable flood detection and response system that monitors rising water levels in real time.

To design a programmable control mechanism that automatically adjusts a physical barrier based on predefined water-level thresholds.

To integrate visual and audible warning indicators that provide immediate alerts during critical flood conditions.

To incorporate a dual-operation feature (Automatic and Manual modes) to enhance system flexibility and user control.

To demonstrate the practical application of embedded systems technology in disaster risk reduction and environmental monitoring.

SYSTEM BLOCK DIAGRAM (IPO)

The Flood Control Barrier System follows the Input–Process–Output (IPO) model. The system receives environmental and user inputs through a water level sensor and push button, processes the data using an Arduino microcontroller, and produces mechanical movement and warning outputs through the servo motor, LCD, LEDs, and buzzer.

INPUT

- Water Level Sensor (Analog A0)
- Push Button (Manual Mode Control)

PROCESS

• Arduino Microcontroller • Water Level

Reading

• Threshold Comparison (Low, Medium, High)

• Mode Detection (AUTO / MANUAL)

• Barrier Position Control Logic

• LED and Buzzer
Activation Logic

• LCD Display Update

OUTPUT

• Servo Motor (Flood Barrier Movement)

• LCD Display

• Green LED (Low Level Indicator)

• Yellow LED (Medium Level Indicator)

• Red LED (High Level Indicator)

• Buzzer

The Input stage consists of a water level sensor that continuously detects the amount of water present and a push button that allows the user to switch between automatic and manual operation modes.

The Process stage is handled by the Arduino microcontroller, which reads the sensor value, compares it with predefined threshold levels, determines the water condition (LOW, MEDIUM, or HIGH), and executes the appropriate control actions for the barrier, LEDs, buzzer, and LCD.

The Output stage includes the servo motor that adjusts the flood barrier position, the LCD that displays the system mode and water level status, the LEDs that indicate the water condition, and the buzzer that activates during high water levels as an alarm signal.

LIST OF COMPONENTS

The system is composed of an Arduino Uno microcontroller that serves as the main control unit of the Flood Control Barrier System. A water level sensor is used to detect the amount of water present in the monitored area. A 16x2 LCD with I2C module is integrated to display the system mode and water level status in real time.

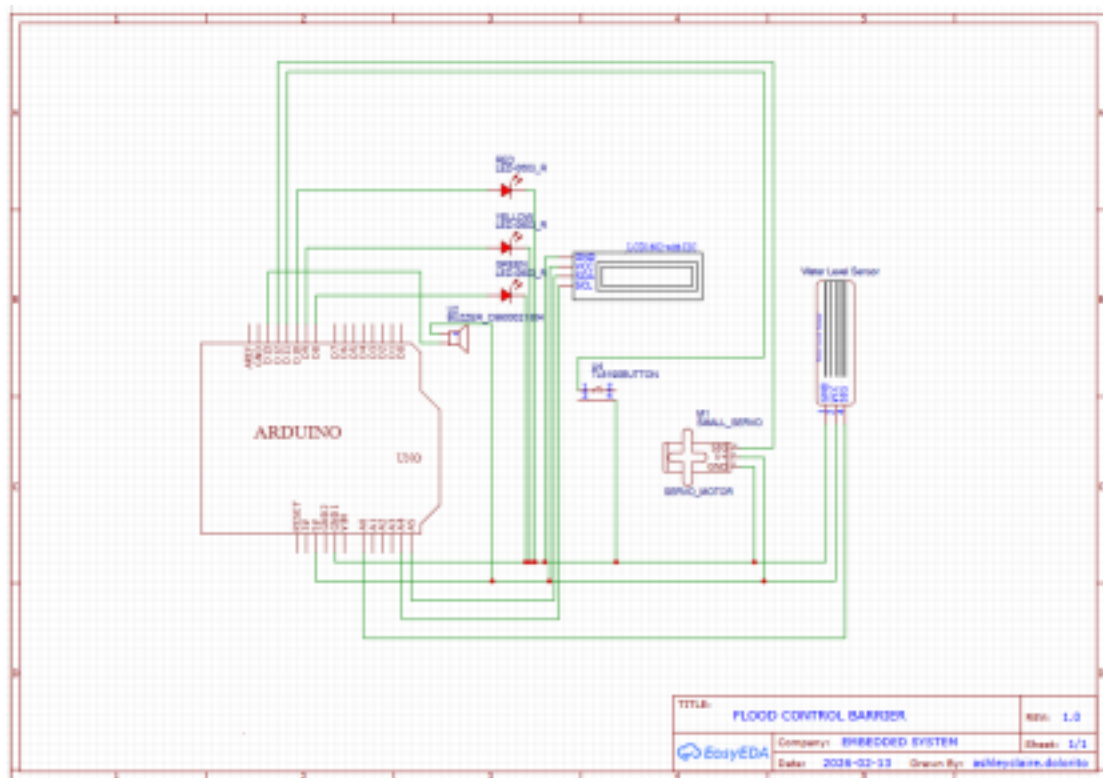
A servo motor is used as the mechanical barrier that moves according to the detected water level. Three LEDs (green, yellow, and red) function as visual indicators to represent low, medium, and high-water levels. An active buzzer is included to provide an audible alarm during critical flood conditions.

A push button is installed to allow switching between automatic and manual modes and to control the barrier position when necessary. Jumper wires are used for circuit connections, while a 5V power supply provides the required voltage for system operation. An enclosure or casing may be used to protect the components and ensure safe installation.

The following are the components that were used during the making of the Flood Control Barrier System:

COMPONENTS	QUANTITY	PURPOSE
Arduino Uno	1	Main microcontroller
Water Level Sensor	1	Detects water level
Servo Motor	1	Controls barrier movement
16x2 LCD with I2C	1	Displays system status
Push Button	1	Manual control input
Green LED	1	Indicates low water level
Yellow LED	1	Indicates medium water level
Red LED	1	Indicates high water level
Buzzer	1	Provides alarm sound
Jumper Wires	MANY WERE USED	Electrical connections
5V Power Supply / USB	1	Supplies power

SCHEMATIC DIAGRAM



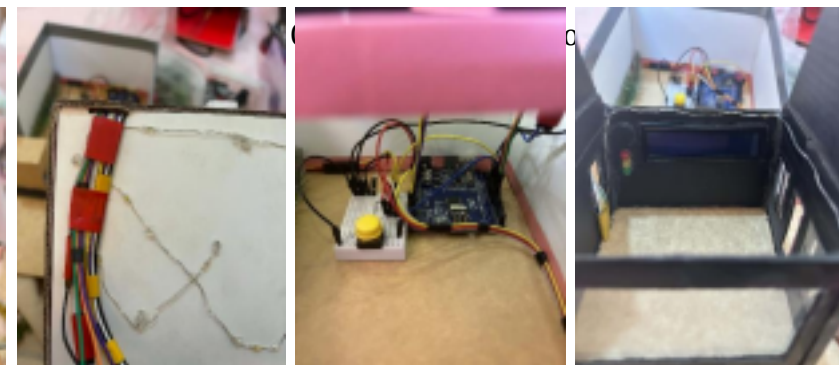
The Flood Control Barrier System is centered around the Arduino Uno, which acts as the main controller, processing inputs from sensors and managing outputs to actuators and alert devices. The system uses a water level sensor connected to Analog Pin A0 to monitor the water in the area, while a push button on Digital Pin 11 allows users to switch between Automatic and Manual modes and control the barrier manually.

The mechanical barrier is operated by a servo motor connected to Digital Pin 12, which adjusts its position to 0°, 45°, or 90° based on the water level. Visual indicators include three LEDs on Digital Pins 8, 9, and 10, showing low, medium, and high-water levels, respectively, while an active buzzer on Digital Pin 13 provides audible alerts during critical conditions.

A 16x2 LCD module connected via the I2C interface displays the system mode and real-time water level, reducing wiring complexity. All components are powered by the Arduino's 5V and GND pins, ensuring stable operation. The schematic effectively integrates the input, processing, and output components, forming a reliable system for monitoring and controlling flood conditions.

Table 1 presents the values used for the linear regression model, including the number of alcoholic drinks consumed (x), GPA/GWA (y), the square of x, and the product of x and y. These values were used to determine the linear relationship between alcohol intake and academic performance.

CIRCUIT SET-UP



PROGRAM CODE

```
#include <Servo.h>
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

Servo barrier;
LiquidCrystal_I2C lcd(0x27, 16, 2);

// Pins
const int sensorPin = A0;
const int greenLED = 8;
const int yellowLED = 9;
const int redLED = 10;
const int buzzer = 13;
const int buttonPin = 11;
const int servoPin = 12;

// Thresholds
int lowLevel = 300;
int mediumLevel = 600;

// State variables
bool manualMode = false;
bool barrierUp = false;
bool lastButtonState = HIGH;

void setup() {
  pinMode(greenLED, OUTPUT);
```

```

pinMode(yellowLED, OUTPUT);
pinMode(redLED, OUTPUT);
pinMode(buzzer, OUTPUT);
pinMode(buttonPin, INPUT_PULLUP);

barrier.attach(servoPin);

lcd.init();
lcd.backlight();

Serial.begin(9600);

lcd.setCursor(0,0);
lcd.print("System Starting");
delay(1500);
lcd.clear();
}

void loop() {

  int waterValue = analogRead(sensorPin);
  bool buttonState = digitalRead(buttonPin);

  // Detect button press (falling edge)
  if (lastButtonState == HIGH && buttonState == LOW) {
    if (!manualMode) {
      // Enter MANUAL mode
      manualMode = true;
      barrierUp = false;
    } else {
      // In manual mode to toggle barrier
      barrierUp = !barrierUp;
    }

    delay(200); // Debounce
  }

  lastButtonState = buttonState;

  Serial.println(waterValue);

  // MANUAL MODE

  if (manualMode) {

    digitalWrite(greenLED, LOW);
    digitalWrite(yellowLED, LOW);
    digitalWrite(redLED, LOW);
    noTone(buzzer);

    lcd.setCursor(0,0);
    lcd.print("Mode: MANUAL ");

    if (barrierUp) {
      barrier.write(90);
      lcd.setCursor(0,1);
      lcd.print("Barrier: UP ");
    } else {
      barrier.write(0);
      lcd.setCursor(0,1);
      lcd.print("Barrier: DOWN ");
    }

    // Long press (2 sec) to return AUTO mode
    if (buttonState == LOW) {

```



```

delay(2000);
if (digitalRead(buttonPin) == LOW) {
  manualMode = false;
  lcd.clear();
}
}

delay(200);
return;
}

// AUTOMATIC MODE

lcd.setCursor(0,0);
lcd.print("Mode: AUTO ");

if (waterValue < lowLevel) {
  barrier.write(0);
  digitalWrite(greenLED, HIGH);
  digitalWrite(yellowLED, LOW);
  digitalWrite(redLED, LOW);
  noTone(buzzer);

  lcd.setCursor(0,1);
  lcd.print("Level: LOW ");

}
else if (waterValue >= lowLevel && waterValue < mediumLevel) {

  barrier.write(45);
  digitalWrite(greenLED, LOW);
  digitalWrite(yellowLED, HIGH);
  digitalWrite(redLED, LOW);
  noTone(buzzer);

  lcd.setCursor(0,1);
  lcd.print("Level: MEDIUM ");
}
else {

  barrier.write(90);
  digitalWrite(greenLED, LOW);
  digitalWrite(yellowLED, LOW);
  digitalWrite(redLED, HIGH);
  tone(buzzer, 1000);

  lcd.setCursor(0,1);
  lcd.print("Level: HIGH ");
}

delay(500);
}

```

MANUAL USER-GUIDE

1. How to Power ON the System

To power on the Flood Control Barrier System, connect the Arduino Uno to a 5V power source using a USB cable. Once powered, the LCD screen will automatically turn on and display:

“System Starting”, which means the system is initializing.

After initialization, the system automatically enters AUTO mode.

2. How Input is Given

The system uses two input devices:

- Water Level Sensor – Automatically detects the water level.
- Push Button – Controls the system mode and barrier position.

Push Button Operations:

- Short Press: Enters MANUAL mode and toggles the barrier UP or DOWN.

Long Press (2 seconds): Returns the system to AUTO mode. Step-by-step input process:

1. When the system is in AUTO mode, the LCD displays: "Mode: AUTO".
2. To enter MANUAL mode, press the push button shortly. The LCD will display: "Mode: MANUAL".
3. In MANUAL mode, each short press toggles the barrier position between DOWN, 45°, and UP.
4. To return to AUTO mode, press and hold the push button for 2 seconds.

3. How the System Responds

AUTO Mode:

Barrier and indicators respond automatically based on water level:

- LOW Level: Barrier stays DOWN (0°), Green LED ON, no buzzer.
- MEDIUM Level: Barrier moves to 45°, Yellow LED ON, no buzzer.
- HIGH Level: Barrier moves to 90° (fully raised), Red LED ON, buzzer sounds.

MANUAL Mode:

- Barrier position can be manually controlled by the user.
- All LEDs and the buzzer remain OFF in manual mode.
- Manual mode overrides automatic barrier operation.

4. Meaning of Outputs (LED, Display, Buzzer)

LCD Display Messages:

- "Mode: AUTO" = System is controlling barrier automatically.

"Mode: MANUAL" = User is manually controlling the barrier. LED

Indicators:

- Green LED = LOW water level
- Yellow LED = MEDIUM water level

- Red LED = HIGH water level

Buzzer:

- Activates only when water level reaches HIGH in AUTO mode, providing an audible warning.

5. Safety Precautions

1. Ensure all wires are properly connected before powering the system.
2. Use only a 5V power supply to avoid damaging the Arduino board.
3. Do not touch exposed wires while the system is powered.
4. Keep the system away from water or moisture.
5. Avoid placing the buzzer close to the ears, as the sound may be loud.

USEFULNESS OF THE SYSTEM

The Flood Control Barrier System is useful because it helps prevent flood damage by automatically monitoring water levels and controlling a mechanical barrier. Flooding can cause property loss, safety hazards, and disruption to daily life. This system provides both visual (LEDs) and audible (buzzer) alerts, making it easier to respond quickly before floodwaters rise to dangerous levels.

Unlike manual flood monitoring, this system works independently once powered on and requires minimal human intervention. It continuously senses water levels and adjusts the barrier automatically in AUTO mode, while also allowing manual control when necessary. This makes it practical and easy to use in homes, small offices, or areas prone to flooding.

Real-Life Application

The system is suitable for areas where water levels need continuous monitoring, such as:

- Homes near rivers or low-lying areas
- Small offices or commercial buildings
- Drainage points or urban flood-prone zones
- Community centers and schools

The system can be installed near entry points of water or along barriers for easy access and monitoring.

Who Can Benefit from It

- Homeowners in flood-prone areas
- Facility managers responsible for building safety
- Local government units monitoring small-scale urban flooding • Caregivers or residents needing early warnings to prevent property or health damage

How It Improves Efficiency, Safety, Automation, and Convenience • Efficiency:

Continuously monitors water levels without the need for constant manual checks.

- Safety: Reduces risk by activating barriers and alerts before flooding becomes critical.
- Automation: AUTO mode adjusts the barrier and triggers warnings automatically.
- Convenience: Simple setup with clear LCD display and push-button control for easy operation.

Possible Future Improvements

Future enhancements could include:

- Integration with a Real-Time Clock (RTC) for scheduled flood checks • Remote monitoring via mobile or web applications
- Multiple barriers control for larger areas
- Rechargeable or backup power supply for uninterrupted operation • Data logging to track water levels and barrier activation history

These improvements would make the system more advanced, reliable, and closer to a full-scale smart flood management solution.